5	**(a)**	When white light is incident on a single slit, a diffraction pattern is formed on a screen. The central fringe of the diffraction pattern is coloured at the edges and has a white central region. Explain this observation.	
		………………………………………………………………………………………………	
		………………………………………………………………………………………………	
		………………………………………………………………………………………………	[2]
		White light consists of <u>all</u> colours of visible light, which is of a <u>continuous range of wavelengths</u>. The <u>longer the wavelength</u>, the <u>greater the degree of diffraction</u>, <u>producing a central fringe of larger width</u>.	**1**
		In the central region where <u>all</u> colours <u>overlap</u>, it is white. **At the edges where not all colours are present, they are coloured.**	**1**
			
		OR White light consists of visible light of a continuous range of wavelengths. **Visible light of every wavelength meet with zero path difference at the centre, (thus meet in phase and undergo constructive interference,) thus all the overlapping colours produces a white region.** **At the edges, there are destructive interference of some light but not for others as the extent of diffraction increases with the wavelength. Hence the edges would not be white but be of the mixed colour of wavelengths that have non-zero intensity.**	
	(b)	**(i)** Explain what is meant by the *Rayleigh criterion* for the resolution of the images of two objects.	
		………………………………………………………………………………………………	
		………………………………………………………………………………………………	
		………………………………………………………………………………………………	[2]
		Two objects will be ***<u>just</u>* <u>seen as separate / *just* distinguishable</u>** when	**1**
		the **<u>first minimum</u> in the <u>diffraction pattern</u> of one image <u>coincides with</u>** the **<u>central maximum of the other</u>**. □	**1**

(ii) The Griffith Observatory in Los Angeles includes an astronomical refracting telescope (Griffith telescope) with an objective lens of diameter 0.305 m.

Calculate the wavelength of light for which the Griffith telescope has a minimum angular resolution of 1.8×10^{-6} rad.

wavelength = …………………………… m [1]

From Rayleigh Criterion,

$\theta_{min} \approx \frac{\lambda}{b}$

$1.8\ x\ 10^{-6} \approx \frac{\lambda}{0.305}$

$\lambda \approx 5.49\ x\ 10^{-7} = \mathbf{5.5\ x\ 10^{-7}\ m}$ **1**

(iii) The asteroid Apophis has a diameter of 325 m. It has been calculated that in the year 2029, its distance of closest approach to the Earth's surface will be 3.0×10^{4} km.

Supporting your answer with calculations, explain whether the Griffith telescope can resolve Apophis.

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

Angular size of asteroid, $\boldsymbol{\theta} \approx \frac{\mathbf{325}}{\mathbf{3.0\ x\ 10^{4} x\ 10^{3}}}$ **1**

$= \mathbf{1.0833\ x\ 10^{-5} = 1.1\ x\ 10^{-5} rad}$

As $\mathbf{1.1\ x\ 10^{-5} rad}$ is greater than $\mathbf{1.8\ x\ 10^{-6}\ rad}$, the angular size of the asteroid exceeds the minimum angular resolution of the Griffith telescope, thus the telescope can resolve the asteroid. **1**

6	In an X-ray tube, electrons are produced from a filament heated by an electric current as shown in